

BUREAU OF INDIAN STANDARDS

DRAFT FOR COMMENTS ONLY

*(Not to be reproduced without permission of BIS or used as an Indian Standard)***DRAFT AMENDEMENT NO. 1 JULY 2026**

TO

IS 19089: 2025**Respiratory Protective Devices — Powered Filtering Devices
Incorporating a Helmet or Hood — Specification**

ICS 13.340.30

Occupational Safety and Health Sectional
Committee, CHD 08Last Date for Comments : **As per the covering letter***[Page 24, Clause 8.8]* — Substitute the following for the existing Clause.**‘8.8 Clogging**

8.8.1 The test equipment and the test atmosphere shall be that described in IS 15322 with the following modifications. At least the filters and the fan shall be in test atmosphere for the test. New filters and a fully charged battery shall be fitted to the device before starting the clogging procedure. The complete device fitted with a fully charged battery and clean filter(s) shall be tested on a standard dummy head connected to a breathing machine adjusted to 30 l/min (20 cycles/min, 1.5 l/stroke, sinusoidal breathing pattern).

8.8.2 Operate the device in a dolomite dust concentration of (C) 400 ± 100 mg/m³ until the required clogging exposure value (K) specified in following table is achieved in time duration (t).

8.8.3 The clogging exposure requirements and example test conditions are as specified in **Annex-C**.

8.8.4 At the end of the test, take the device out of the dust chamber, clean on the outside if necessary and test for flow rate in accordance with **8.7** and for the penetration requirements at this flow rate in accordance with **8.12.2**, except:

- a) That conditioning is not required; and
- b) Only the penetration test of equivalent IS shall be performed.’

[Page 36, Clause 9.6] — Insert the following new clause after 9.6.*[Page 28, Clause – 8.12.3.2, Para 3]* — Substitute ‘value’ for ‘valve’.**‘9.7 BIS Certification Marking**

The product(s) conforming to the requirements of this standard may be certified as per the Conformity Assessment schemes under the provisions of the Bureau of Indian Standards Act, 2016 and the Rules and Regulations framed thereunder, and the products may be marked with the Standard Mark.’

[Page 41, ANNEX B] — Add the following after ANNEX B

‘ANNEX C

(Clause 8.8)

Clogging Exposure Requirements and Example Test Conditions

Filter Classification	Required Clogging Exposure K (mg·h/m ³)
a) Particle filters only	
TH1P	400
TH2P	400
TH3P	200
b) Gas/vapour filters only	
	No clogging requirement
c) Combined filters	
TH1(Gas)P	200
TH2(Gas)P	200
TH3(Gas)P	100
NOTE: Gas = gas component of the classification (Refer Table1).	
d) Special filters	
Special filters (If combined with Particulate filter)	100

The clogging exposure shall be determined by maintaining a constant product of dust concentration and test duration:

$$C \times t = K$$

where

C = measured influent dolomite dust concentration of the test atmosphere (mg/m³);

t = test duration (h); and

K = required clogging exposure value for the relevant filter classification (mg·h/m³).

Any combination of dust concentration and test duration may be used, provided that:

a) the product $C \times t$ equals the applicable value of K ; and

b) the measured dust concentration C is within the specified test atmosphere concentration range of (400 ± 100) mg/m³.

NOTE Examples of equivalent test conditions are given below:

- For TH1P and TH2P filters ($K = 400 \text{ mg}\cdot\text{h}/\text{m}^3$), testing may be conducted at $C = 400 \text{ mg}/\text{m}^3$ for 1.0 h, $300 \text{ mg}/\text{m}^3$ for $t = 1.33 \text{ h}$, or if $C = 500 \text{ mg}/\text{m}^3$ for $t = 0.80 \text{ h}$
- For TH1(Gas)P, TH2(Gas)P and TH3P filters ($K = 200 \text{ mg}\cdot\text{h}/\text{m}^3$), testing may be conducted at $C = 400 \text{ mg}/\text{m}^3$ for $t = 0.5 \text{ h}$, $C = 300 \text{ mg}/\text{m}^3$ for $t = 0.67 \text{ h}$, or $C = 500 \text{ mg}/\text{m}^3$ for $t = 0.40 \text{ h}$.
- For TH3(Gas)P and special filters ($K = 100 \text{ mg}\cdot\text{h}/\text{m}^3$), testing may be conducted at $C = 400 \text{ mg}/\text{m}^3$ for $t = 0.25 \text{ h}$, $C = 300 \text{ mg}/\text{m}^3$ for $t = 0.33 \text{ h}$, or or $C = 500 \text{ mg}/\text{m}^3$ for $t = 0.2 \text{ h}$

For all filter classifications, the test duration shall be adjusted such that the product of dust concentration and test duration equals the applicable clogging exposure value (K).'

(CHD 8)